

“CubeRT-02”
CETYS University –
StellarNet



Problem Description

- As of now, nanosatellites are the backbone of various industries such as communication, weather monitoring, navigation and aerospace. As of now, most satellites rely on quickly sharing data back and forth with a ground station. Which renders impossible quick response times, diminishes the quantity of tasks that can be done and renders complex tasks far away from earth impossible.
- The lack of proven flight missions using AI in their control algorithms severely limits the possibilities considered by designers. Increasing costs, limiting the scope of what can be accomplish and decreasing the reach of these technologies.



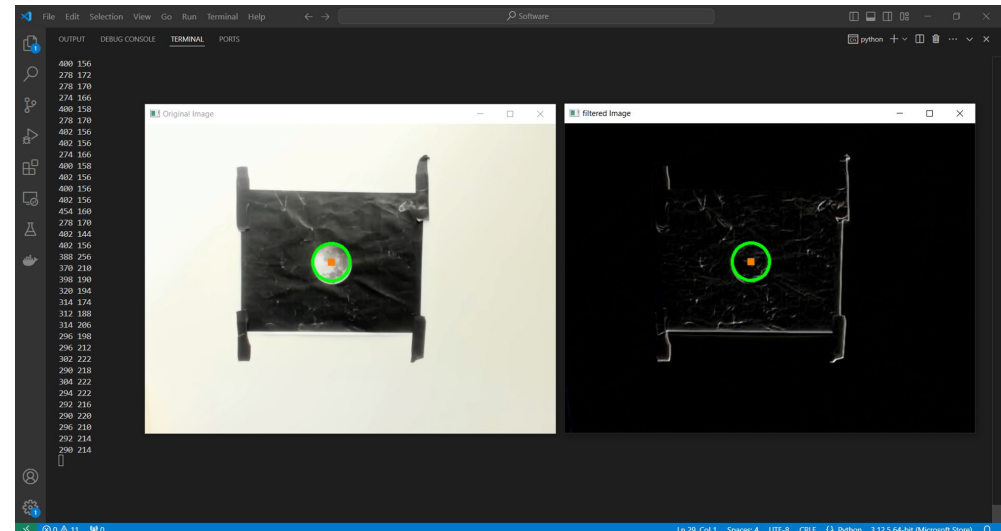
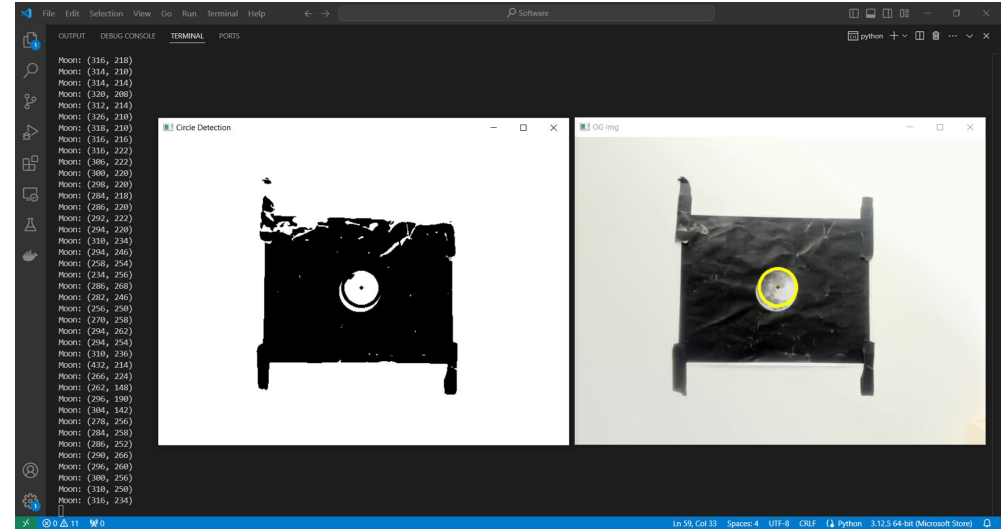
Solution Design

Detection:

- *Thresholding*
- *Sobel Convolution Filters*
- Machine Learning Models (CNN)
- *Yolov8 Object Detection Model*

Control:

- Object Detection
- PID control system



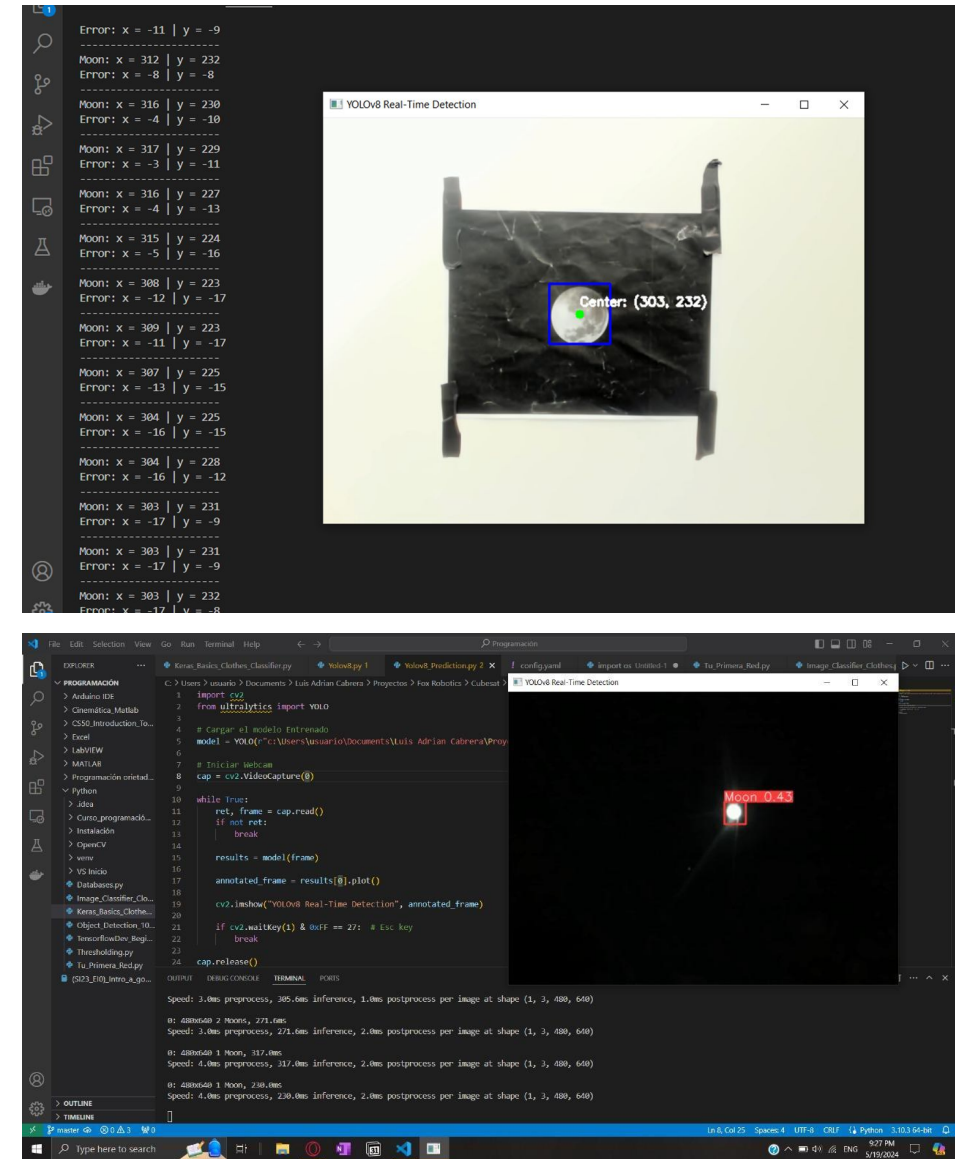
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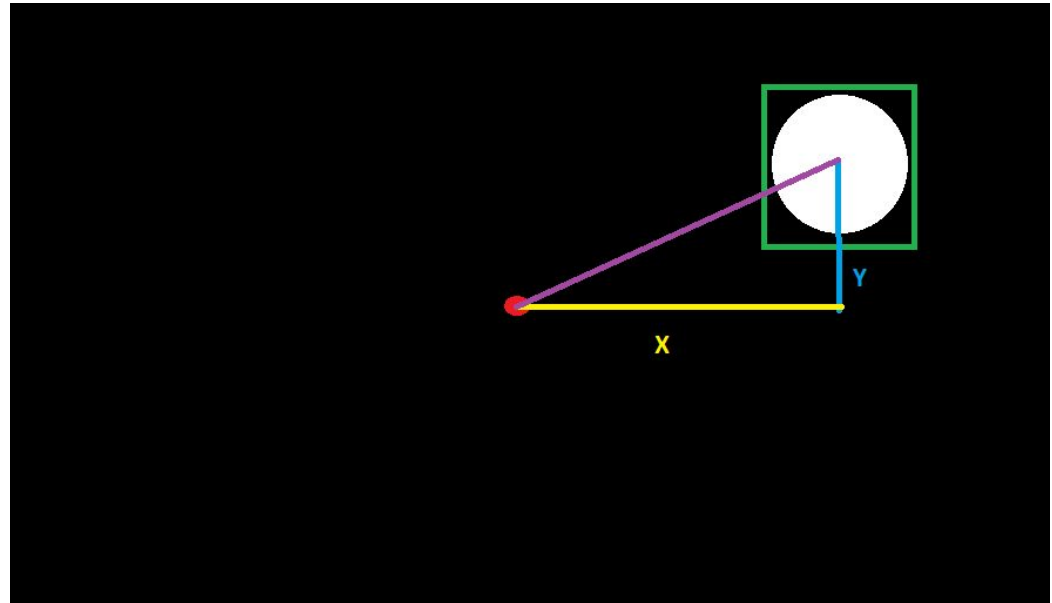
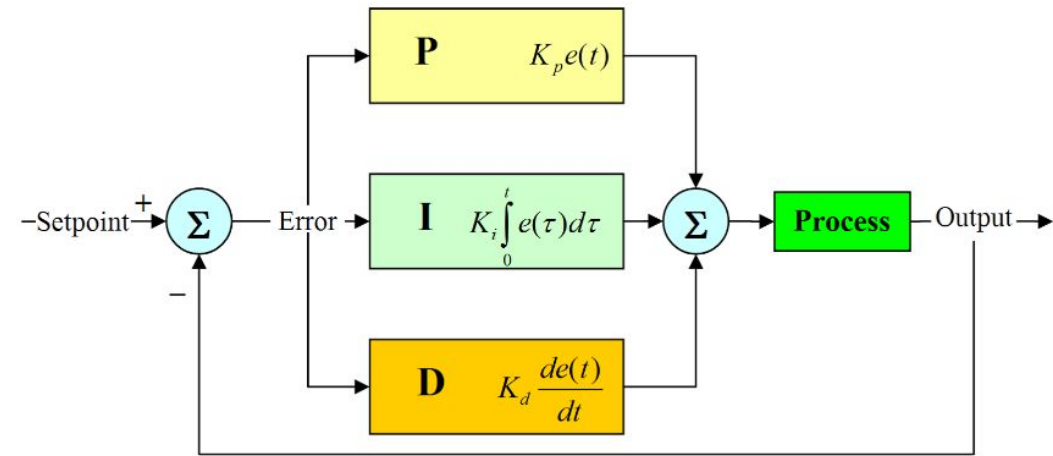
Solution Design

Detection:

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Control:

- *Object Detection*
- *PID control system*



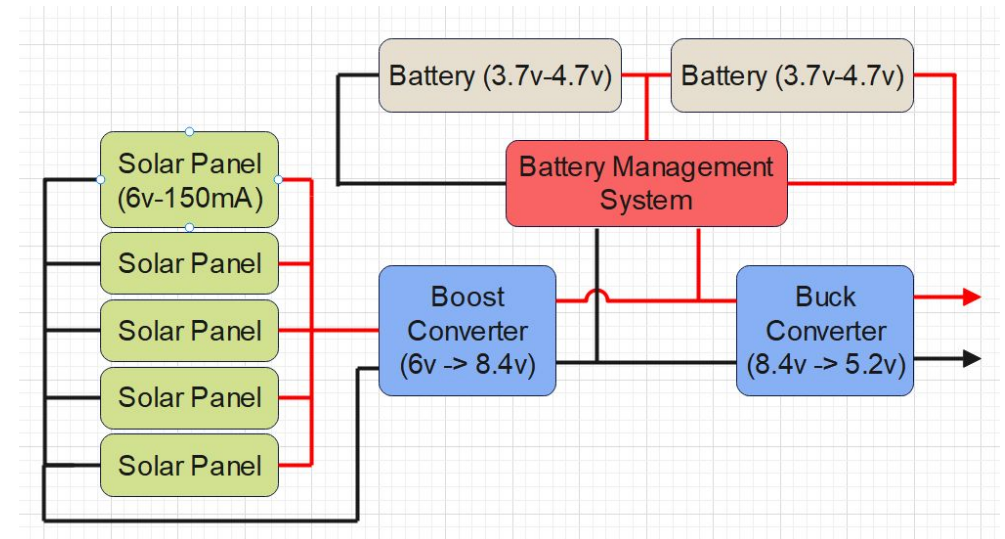
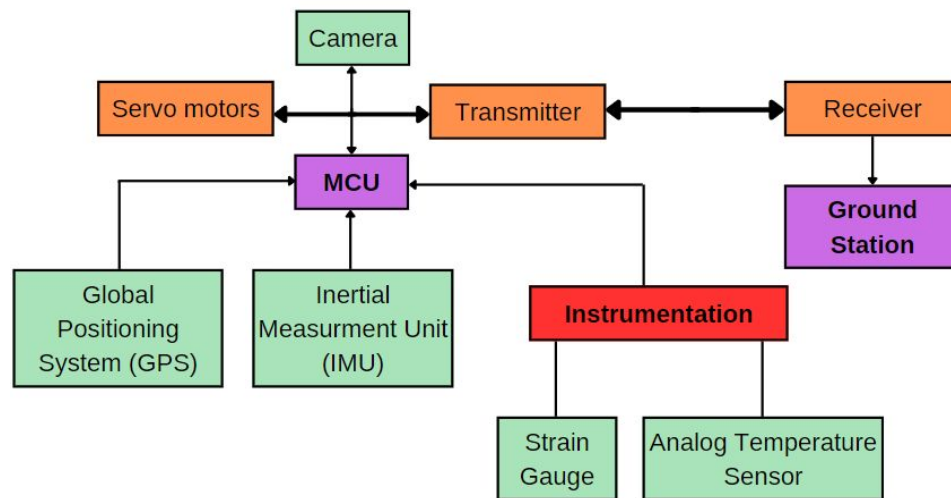
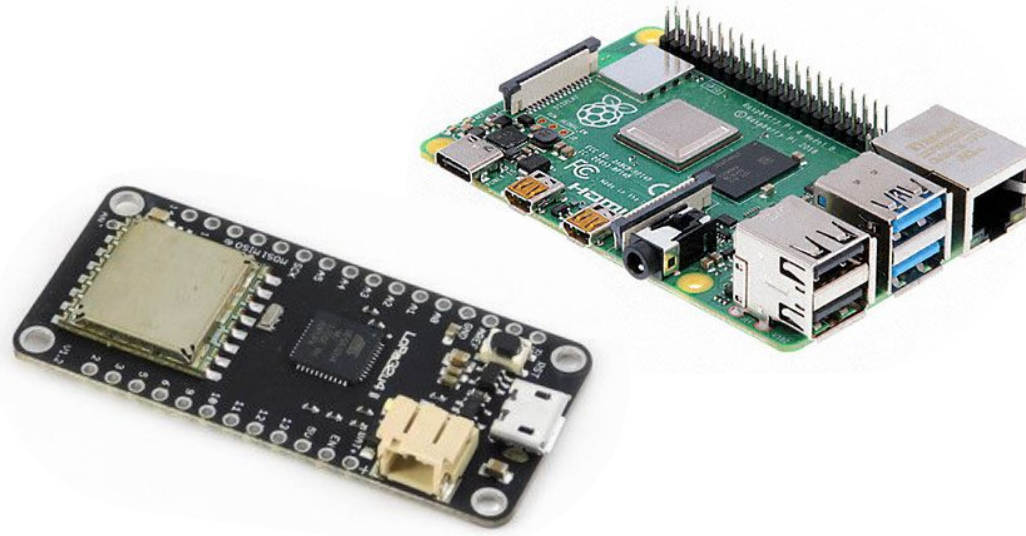
Technology Selection

Hardware:

- Raspberry PI 4B + Pi-Camera
- LoRa 32u4 II

Used Sensors

- Inertial Measurement Unit MPU-6050
- GPS NEO6MV2
- Temperature Sensor LM35
- Strain Gauges for deformation



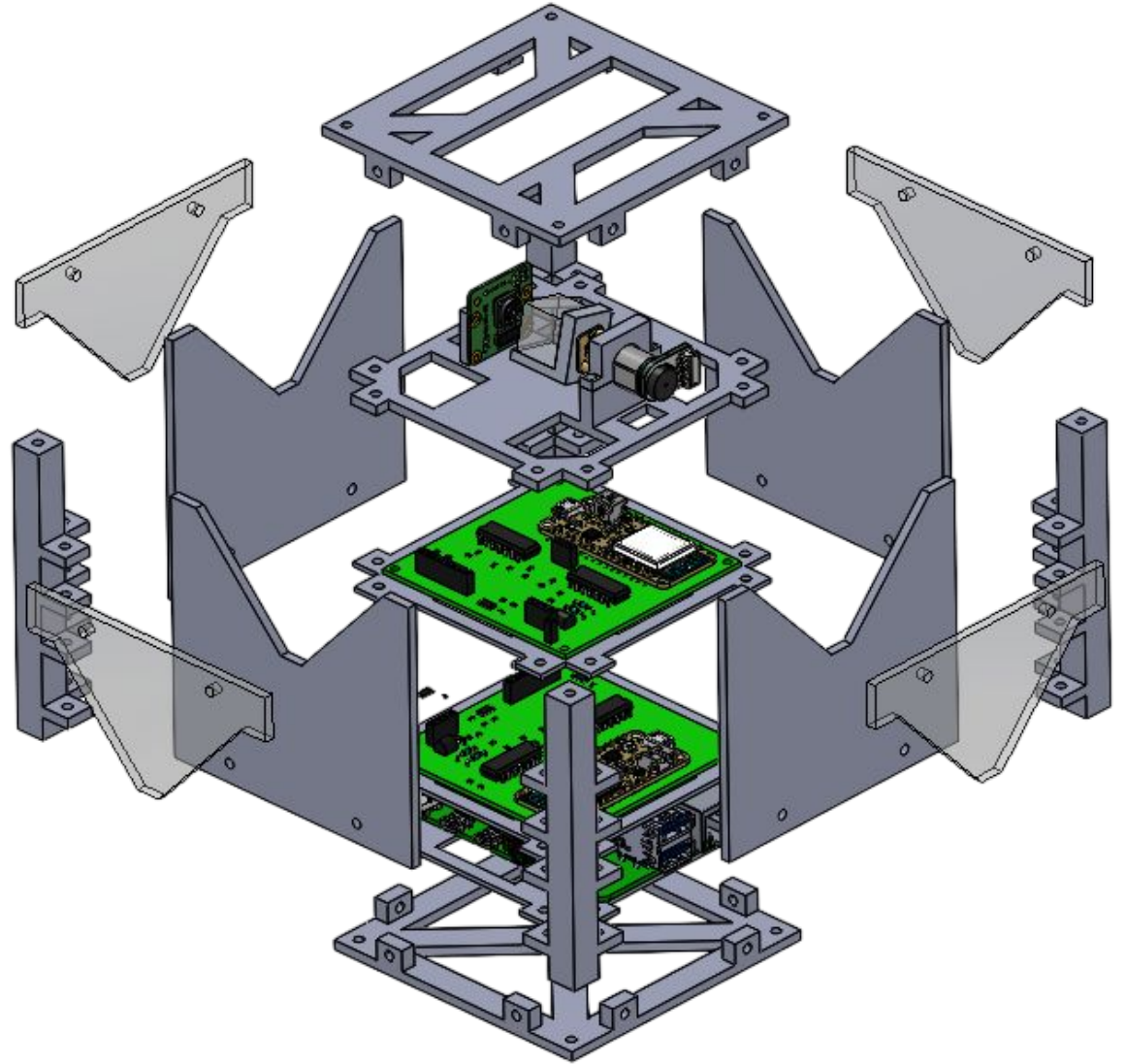
Technology Selection

Optical system

- Gear motor
- Beam splitter prism

Structural design

- Design validation via random vibration and transient structural finite element analysis
- CAD software



Technology Selection

- Ubuntu Mate 22.04 OS + YOLOv8x Architecture
- **Object Storage Service (OBS) | OBS Browser+**
- **Huawei ModelArts -> ExeML**

10k Objects = Local Source + Data Augmentation

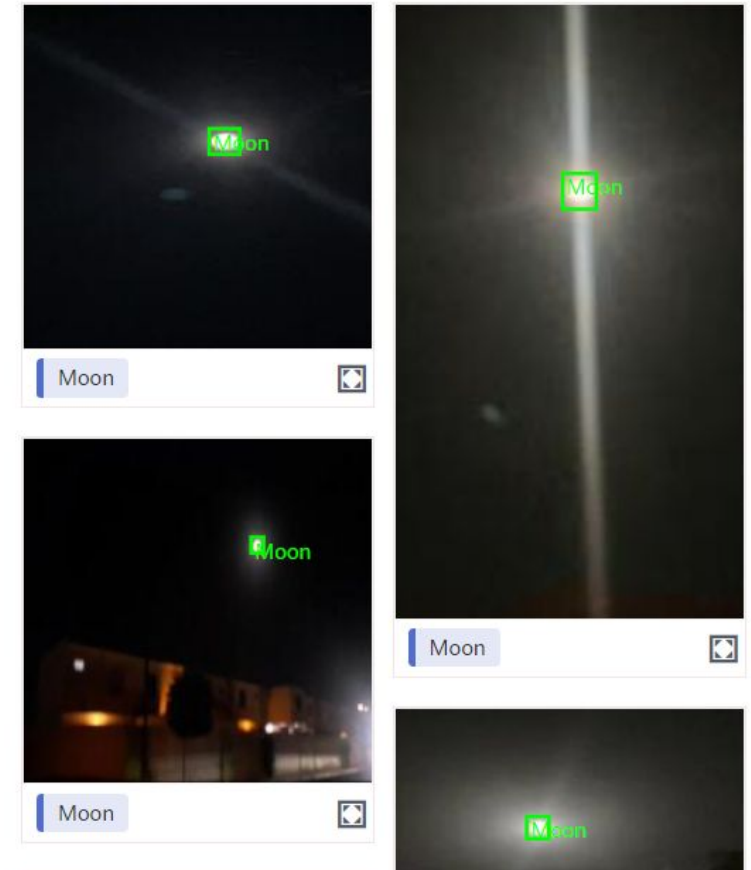
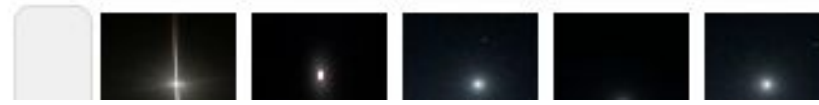
Mean Average Precision (mAP) ? 0.8874

Prediction Results ?



Mean Average Precision (mAP) ? 0.9913

Prediction Results ?



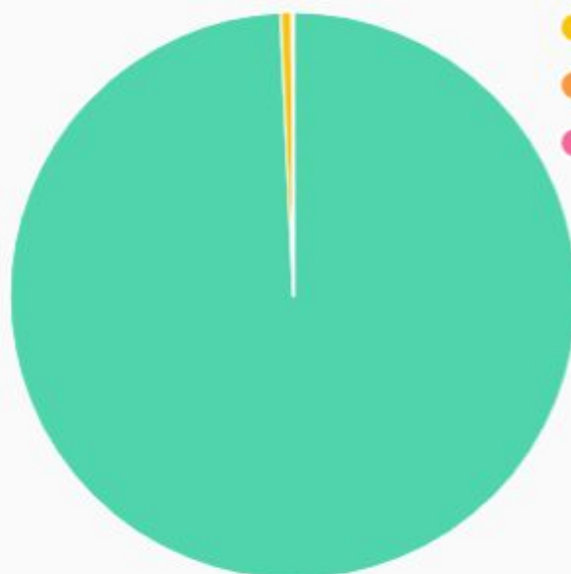
Average Precision (AP) ?

Classes	Moon
Average precision	0.9913

False Positive Analysis ?

total ▼

total

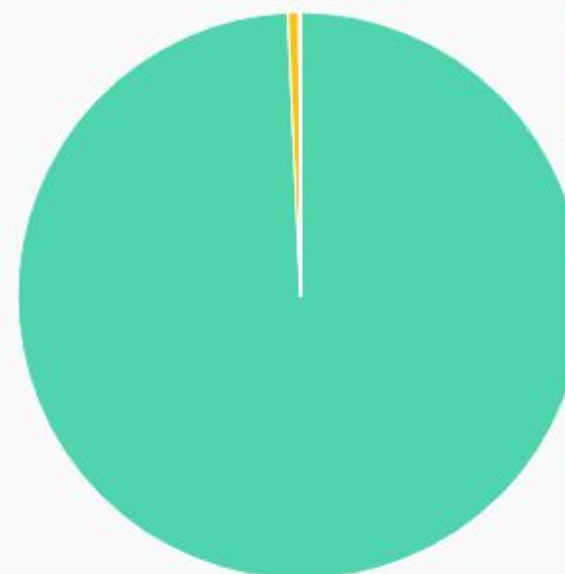


- True positive
- Position false negative
- Class false negative
- Background false negative

False Negative Analysis ?

total ▼

total



- True positive
- Position false negative
- Class false negative
- Background false negative

Problem #1



provider. The cloud service provider does

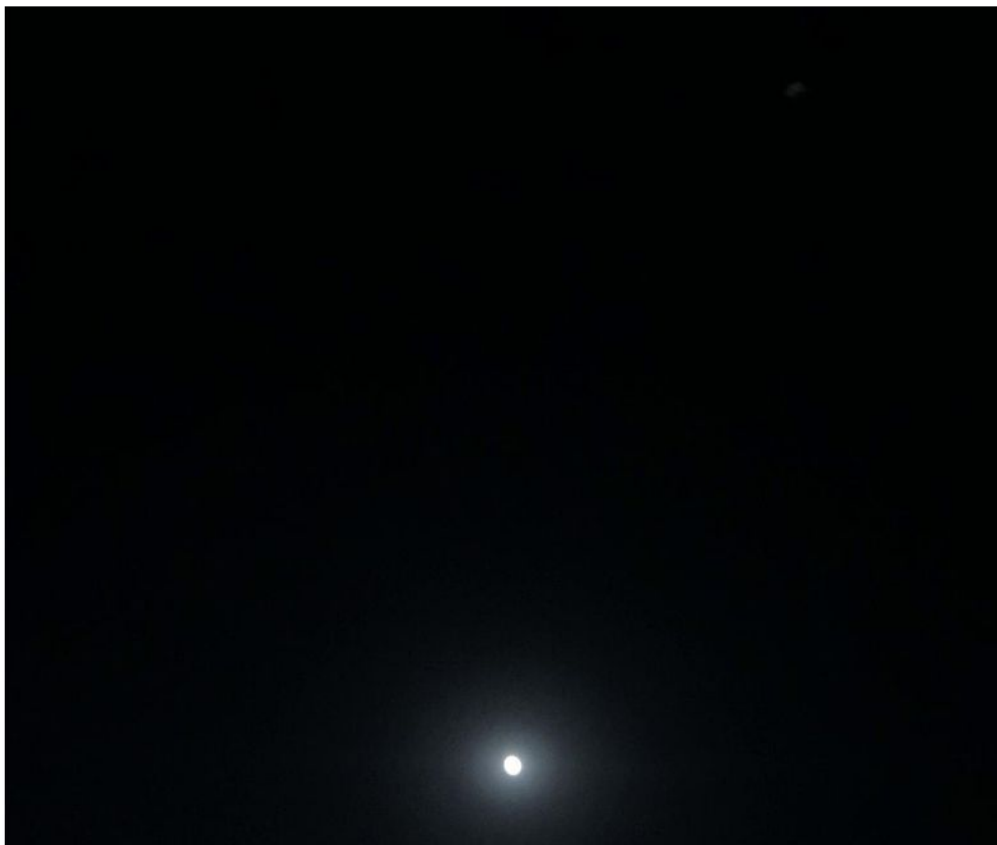
atelier_automl_upredict_success

```
1 {  
2   "detection_classes": [  
3     "Moon"  
4   ],  
5   "detection_boxes": [  
6     [  
7       101,  
8       359,  
9       312,  
10      512  
11     ]  
12   ],  
13   "detection_scores": [  
14     0.9476944208145142  
15   ]  
16 }
```

Generated by AI Cloud
User ID: 8

Problem #1

Test Image Preview



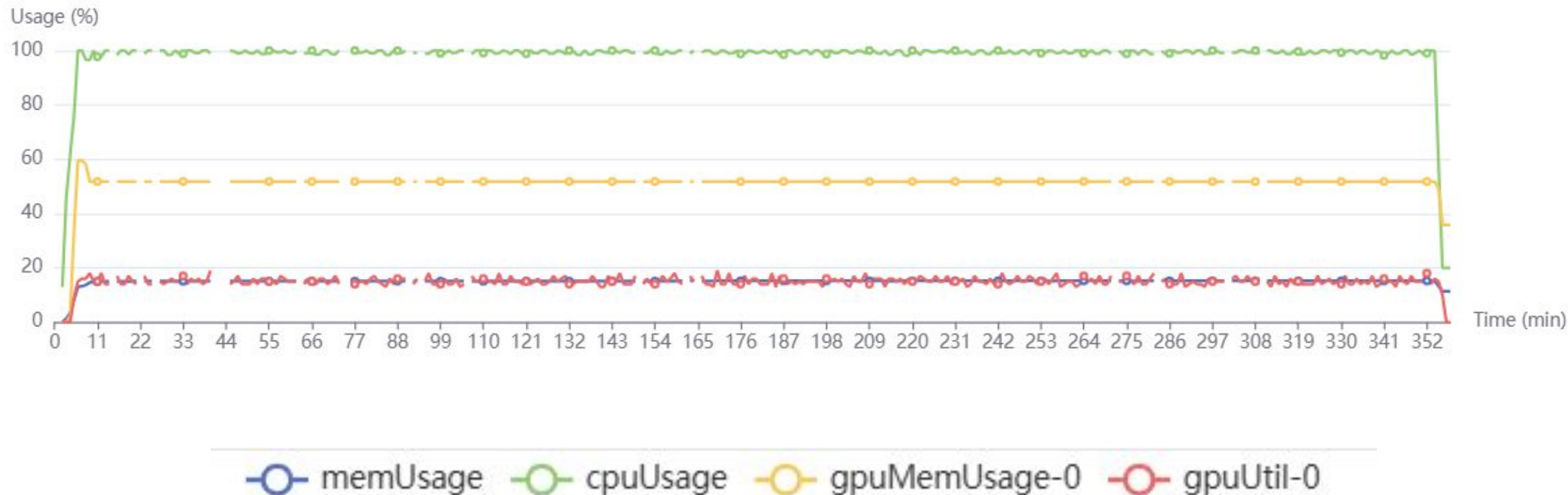
Test Result The inference service model is deployed and provided by you. The inference result is generated by AI and does not represent the opinion of the cloud service provider. The cloud service provider does not provide any guarantee.

[atelier_automl_upredict_failure](#)

Check the [logsatelier_webservice_or](#) or [view the help information](#) Predict



Problem #2



Problem #3

[Usage Guides](#)[Prediction](#)[Instance](#)[Configuration Updates](#)[Monitoring](#)[Events](#)[Logs](#)[Cloud Shell](#)[Tags](#)

API URL <https://infer-modelarts-ap-southeast-1.modelarts-infer.com/v1/infers/a03f8cca-f519...>

Model

Parameter Configuration

^ **POST** /

Input Parameter

Name	Type
—	—

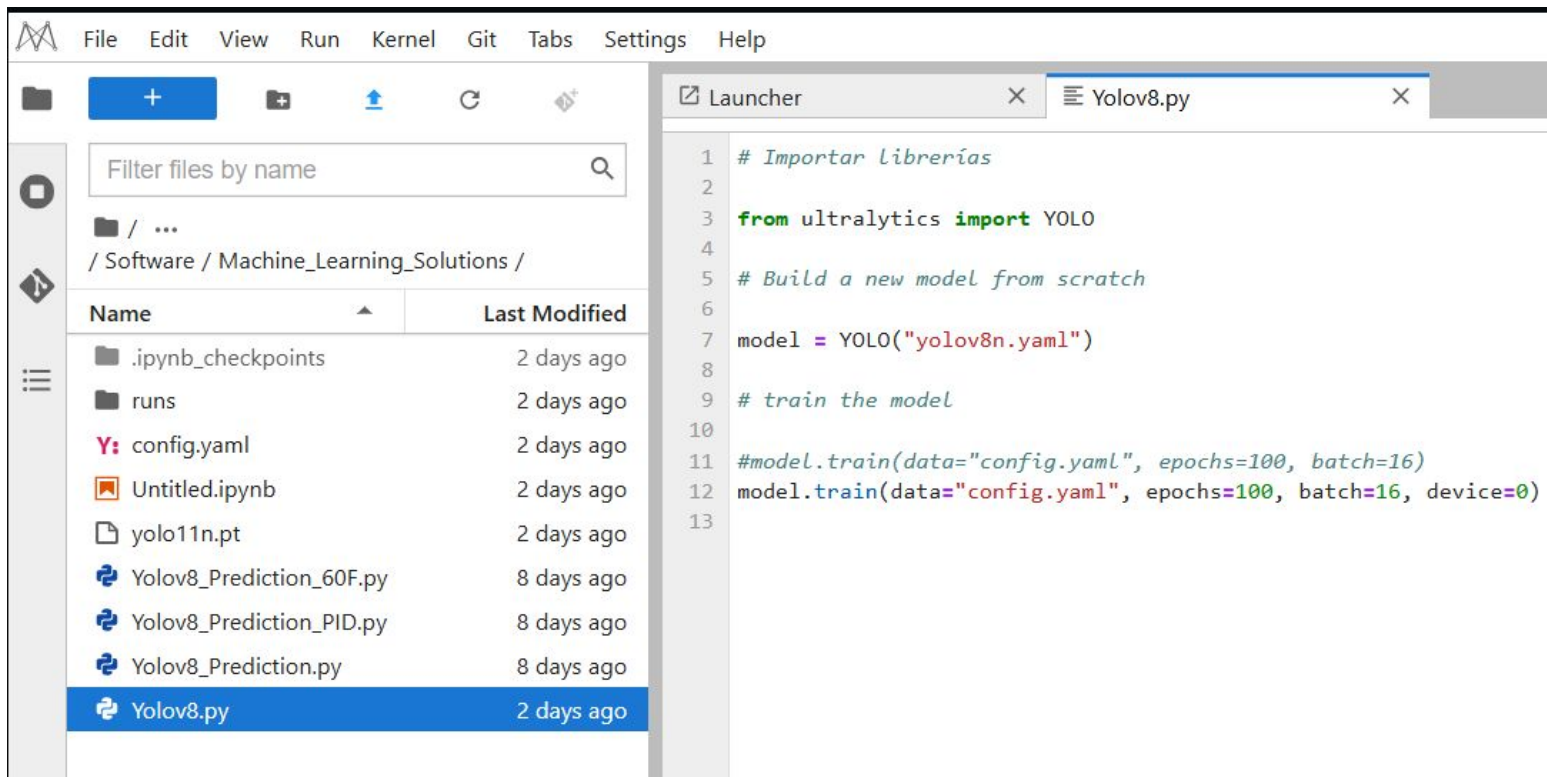
Output Parameter

Name	Type
—	—



Technology Selection

- Ubuntu Mate 22.04 OS + Yolov8x Architecture
- ***Object Storage Service (OBS) | OBS Browser+***
- ***Huawei ModelArts -> JupyterLab: Notebooks***
- ***GPU Support (CUDA)***



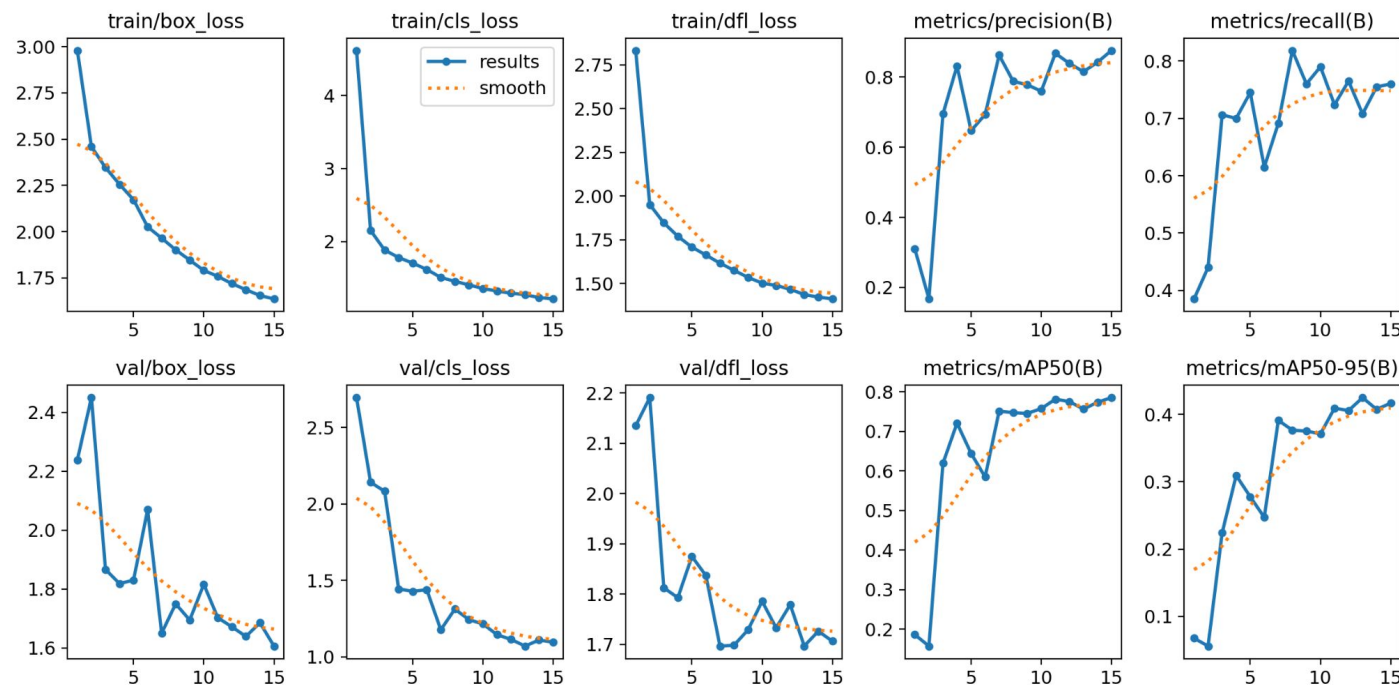
Technology Selection

Local Hardware

- Intel(R) Core i7-4810 2.80GHz
- Intel(R) HD Graphics 4600
- NVIDIA Quadro K2100M

Results:

- Precision: 87.6%
- Recall: 76%
- mAP50: 78.5%
- mAP50-95: 41.8%



Training Time: 70 hours

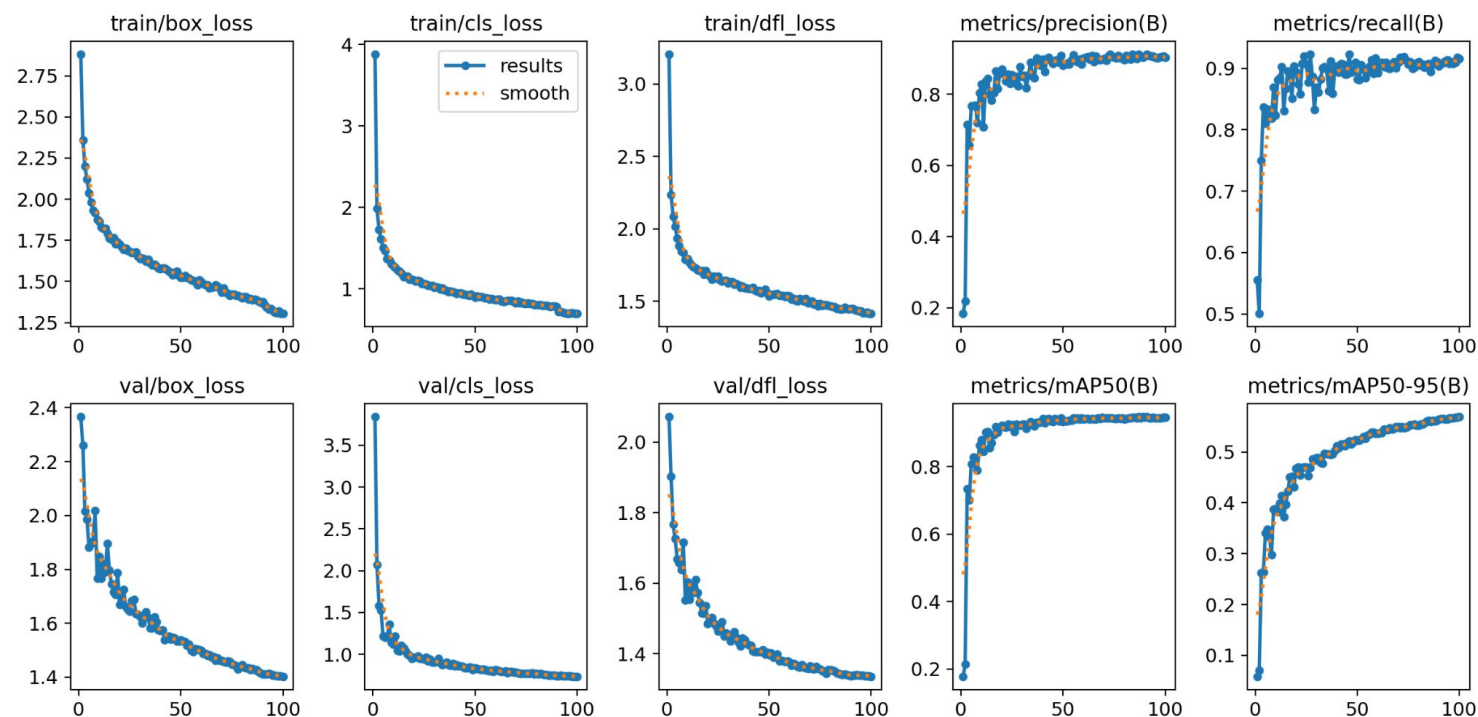
Technology Selection

Cloud Notebook

- GPU: 1*Vnt1(16GB)|CPU: 8vCPUs 64GB
- conda3-cuda10.2-cudnn7-ubuntu18.04

Results:

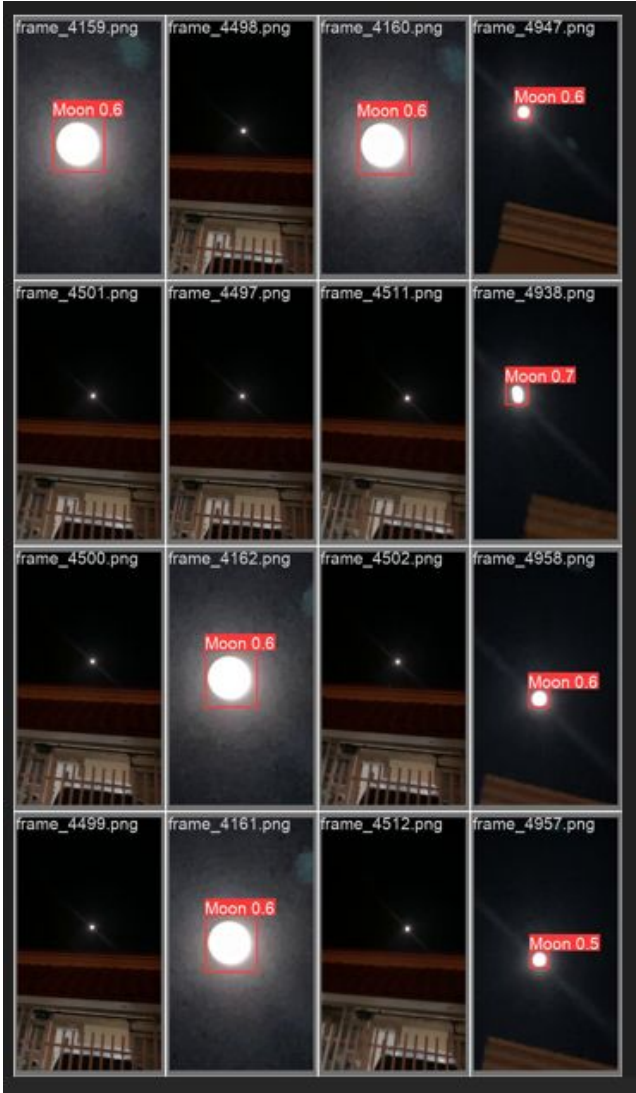
- Precision: 90.3%
- Recall: 91.59%
- mAP50: 94.55%
- mAP50-95: 58.92%



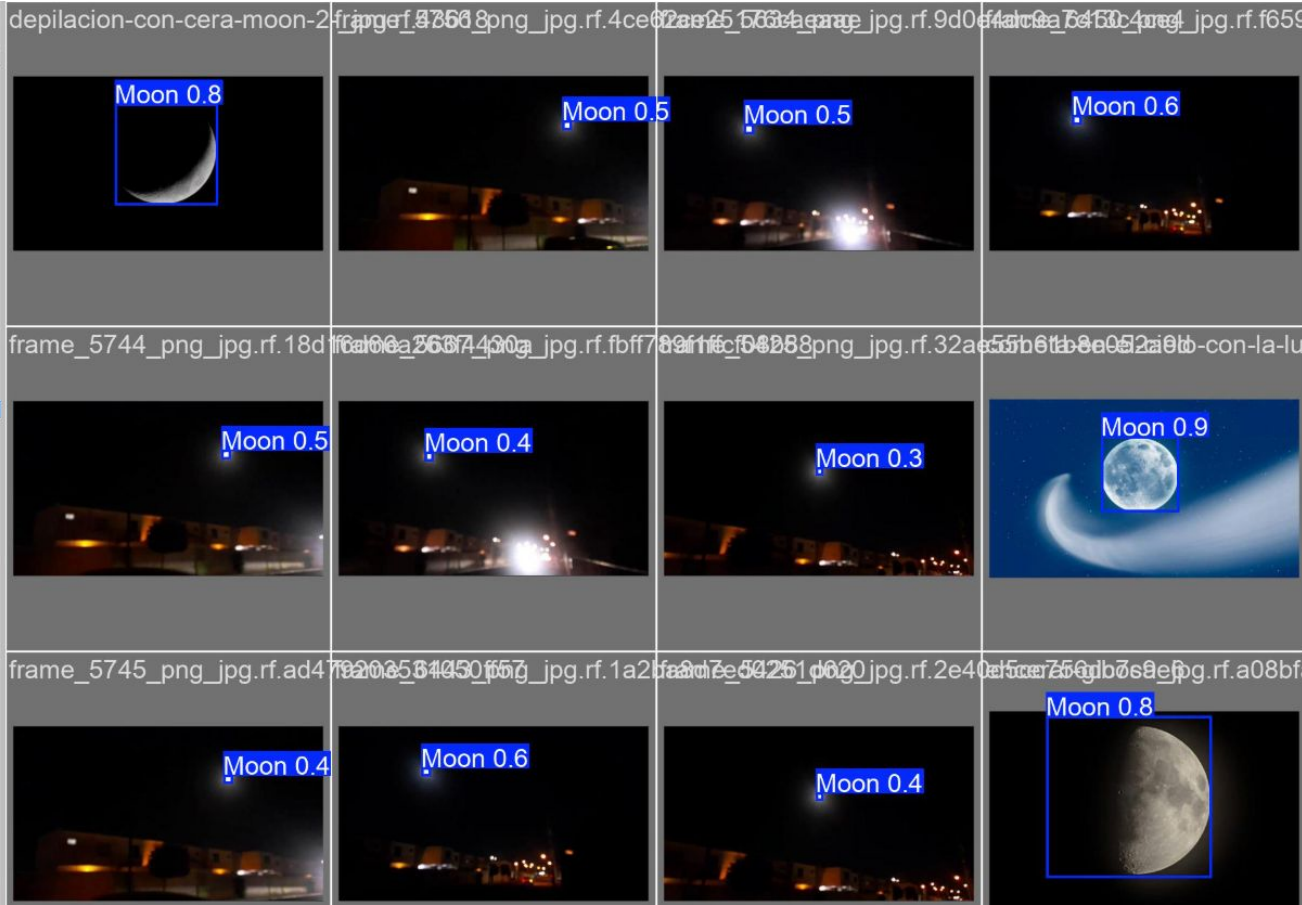
Training Time: 2 hours

Results

Local Training

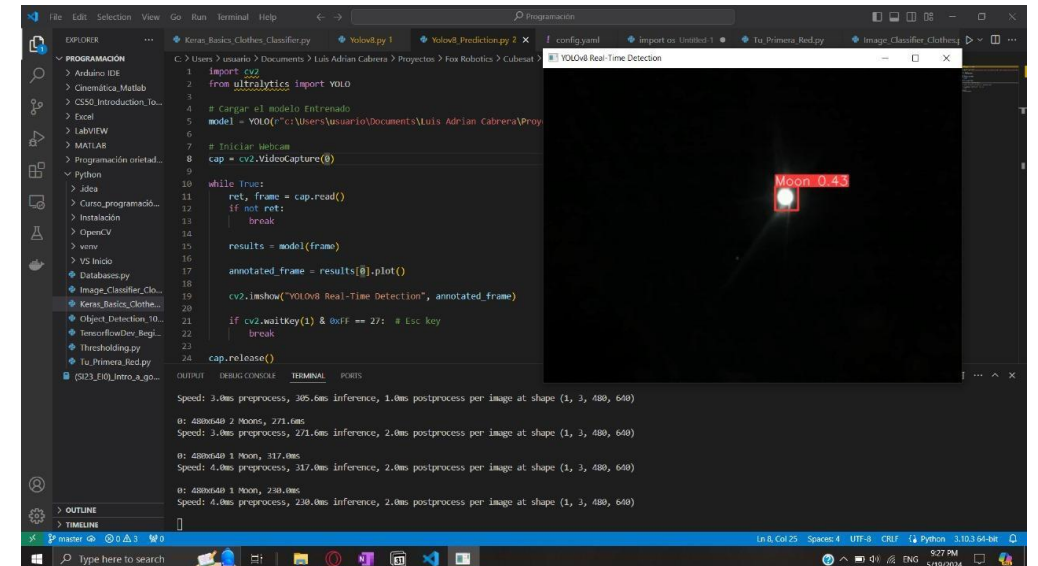
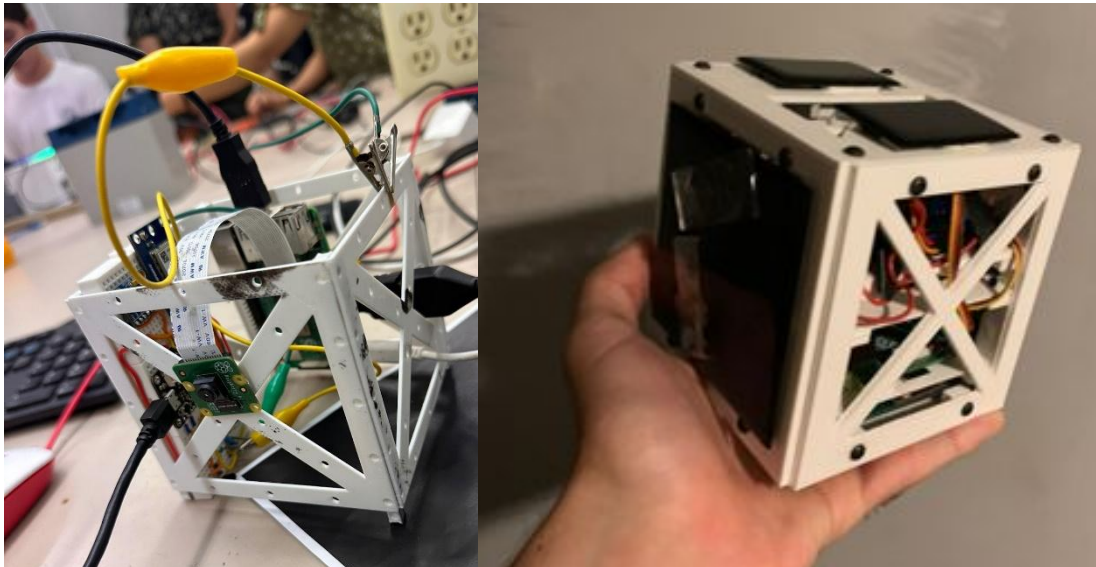


Cloud Training

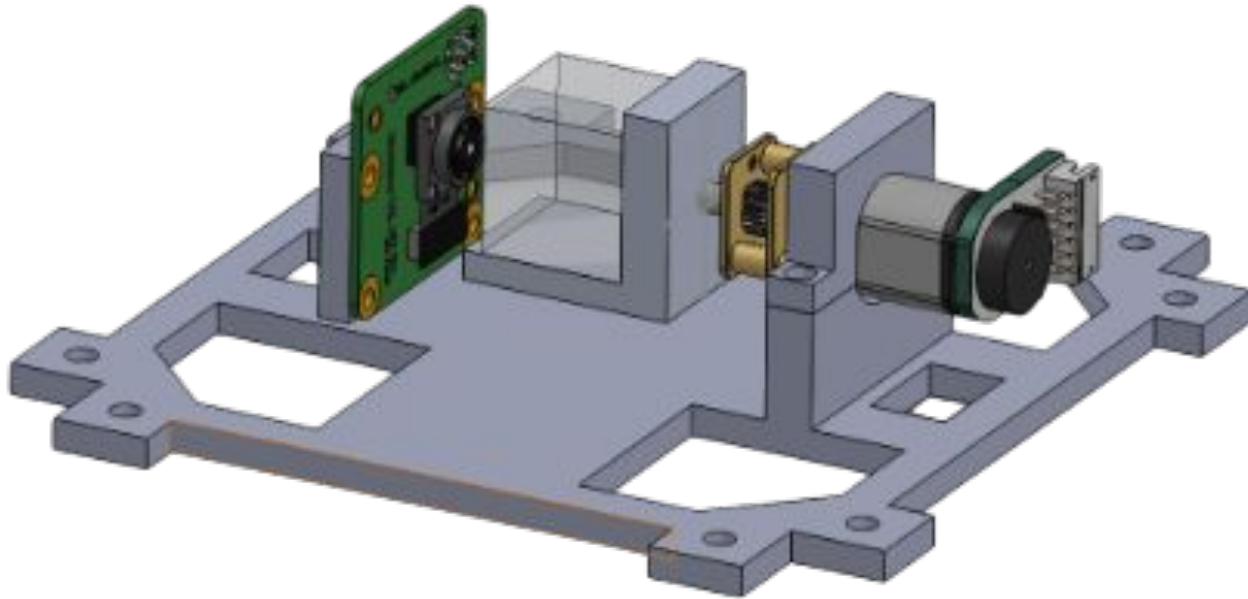


Implementation

The CubeRT-02 is a cutting-edge microsatellite prototype designed to transform the way scientists utilize AI in space exploration. Leveraging the power of Cloud Computing and the ModelArts platform, extensive training iterations were conducted to develop a highly refined and precise control system. This system is capable of seamlessly detecting and tracking Earth's natural satellite, ensuring accurate and reliable performance in space missions.



Implementation



Optical system design criteria

- The captive element (camera) must be protected from direct sunlight as it would burn the system.
- Any circular motion actuator must be able to generate a back and forth motion to annulate any generated momentum.
- All internal mechanisms must not require the displacement of any wire or any other component that may cause a critical failure due to disconnection or unbounding.

Implementation

Area	Component	Price (Dlls)	Total/Area
Electronic	Raspberry 4B + Pi Cam	\$ 85.00	\$ 250.00
	2 x Lora 32u4 II	\$ 100.00	
	PCB Manufacture	\$ 15.00	
	Various electronic pieces	\$ 10.00	
	Modules (MPU, GPS, etc)	\$ 20.00	
	BMS + Solar Panels	\$ 20.00	
Mechanical	3D Prints	\$ 10.00	\$ 15.00
	Screws & bolts	\$ 5.00	
Software	ModelArts platform	\$ 85.00	\$ 85.00



Effect & Value

Proving that IA is not only ready for space but that it is also our best way to improve how we manage our satellites means that in the near future, challenges we deemed impossible will be reachable. Better, faster and more precise information can change science forever and even save lives.

This research aims to be part of the foundation of the further development of other satellite-based tools, as by pushing onward the development of this technology, there is no other limit than the sky



Thank you.

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每个组织，构建万物互联的智能世界。
Bring digital to every person, home and
organization for a fully connected,
intelligent world.

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